Case Study: Revolutionizing EV Motor Efficiency Through Rare-Earth Material Reduction

Client: Automotive OEMs & Global EV Propulsion Manufacturers

Prepared by: FEAAM GmbH

Subject: Reducing Rare-Earth Magnet Dependency in EV Motors via Harmonic Utilization Topology

Executive Summary

As the electric vehicle (EV) market expands, the heavy reliance on rare-earth materials like Neodymium and Dysprosium presents a critical risk to supply chain stability and production costs. FEAAM GmbH has engineered a structural solution: a Harmonic Utilization Topology that extracts more power from significantly less magnet mass. By redesigning the rotor core to utilize the 3rd magnetic harmonic as a "working wave," FEAAM achieves identical performance to a benchmark EV motor while reducing magnet weight by 34%.

The Strategic Challenge: The Rare-Earth Bottleneck

While Permanent Magnet Synchronous Machines (PMSMs) are the industry standard for their high torque density, they suffer from two major vulnerabilities:

- Supply Chain Risk: Volatile rare-earth prices create high-risk exposure for high-volume OEMs.
- Performance Compromises: Standard alternatives like induction motors often result in lower performance or higher complexity.
- Parasitic Wasted Energy: Conventional rotors generate "magnetic noise" (sub-harmonics) that create localized heat and energy loss rather than useful torque.

The FEAAM Innovation: Harmonic Utilization Topology

FEAAM's approach is a "drop-in" structural reimagining of the rotor that requires no changes to the existing stator or winding data.

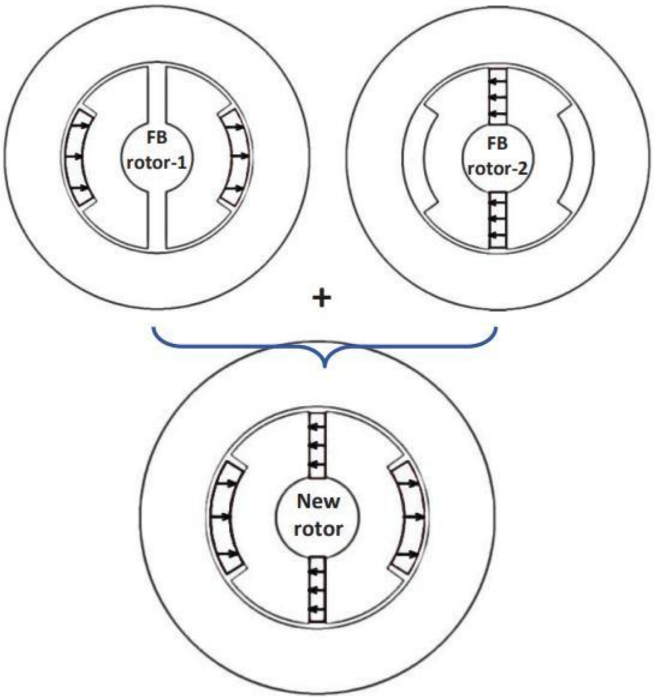

Fig. 1. New rotor design as combination of FB rotor-1 and FB rotor-2.

How it Works:

- Dual-Topology Integration: Combines lateral-magnet and spoke-magnet architectures into a single core.
- Selective Flux Barriers: Air-filled barriers increase resistance for unwanted waves while amplifying the 3rd harmonic to drive the motor.
- Sub-Harmonic Cancellation: By using oppositely magnetized segments, FEAAM achieves a 180-degree phase shift in sub-harmonics, resulting in their complete cancellation.
- Thermal Protection: This cancellation significantly reduces rotor heating, protecting magnets from thermal damage and simplifying cooling.

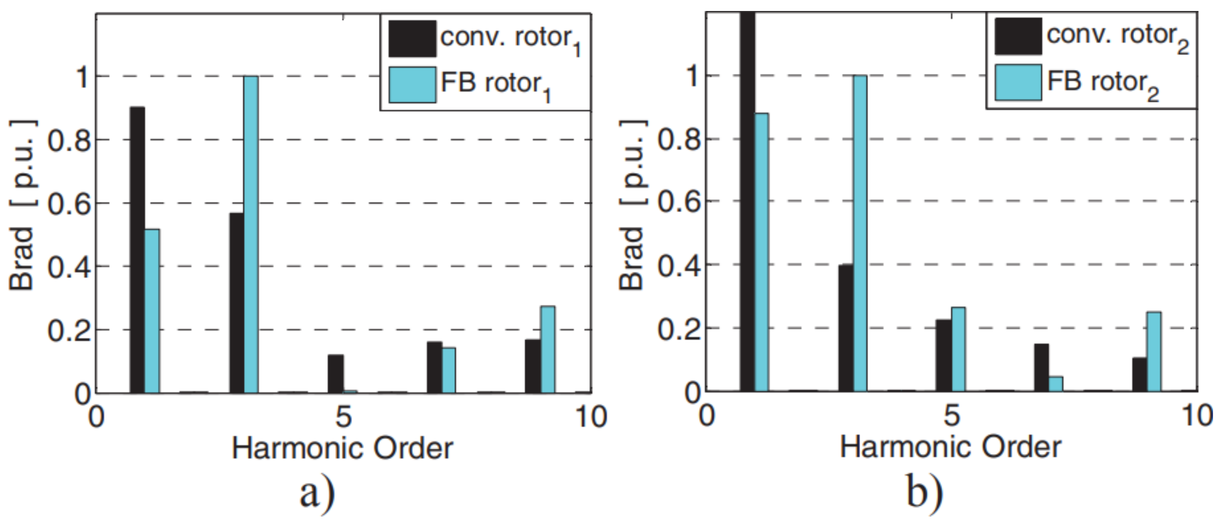

Fig. 2. Comparison of air-gap flux density harmonics due to rotor field, a). Rotor-1, b). Rotor-2.

Benchmarking: FEAAM vs. BMW-i3 Traction Motor

To prove the technology's effectiveness, FEAAM optimized the rotor of a BMW-i3 motor while keeping all other specifications identical.

Metric	**Reference BMW-i3**	**FEAAM Optimized Design**	**Business Impact**
Magnet Weight	1.90 kg	1.25 kg	34% Material Saving
PM Flux-linkage	44 mVs	60 mVs	36% More Flux Produced
Flux Utilization	23.2 mVs/kg	48.0 mVs/kg	106% Efficiency Jump
Torque Utilization	158 Nm/kg	240 Nm/kg	52% Better Torque Use

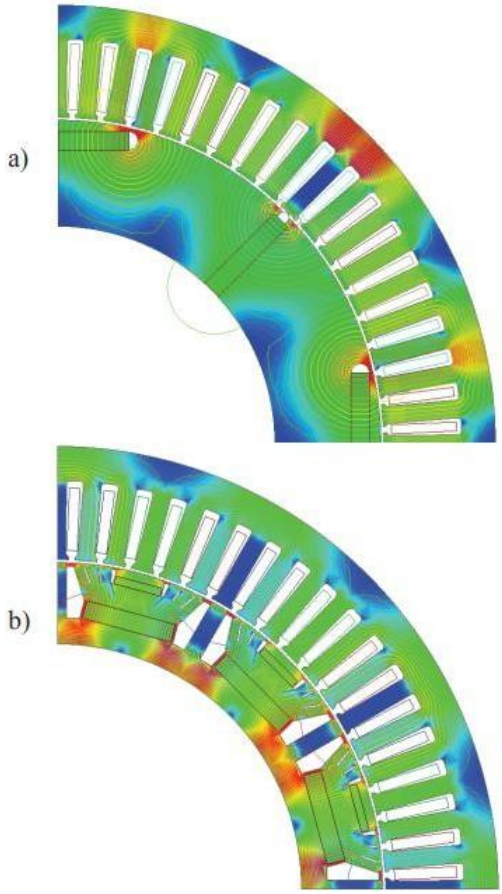

Fig. 3. Study Traction PM Machines; a). New PM Rotor, b). BMW-i3 Rotor.

The Bottom Line for Business

The Harmonic Utilization Topology provides a massive competitive advantage by decoupling success from rare-earth market volatility.

- **Direct Cost Reduction:** Saving 0.65 kg of magnet material per motor represents millions in annual savings for high-volume production.
- **Performance in City Driving:** At lower excitation currents typical of urban cycles, the FEAAM design produces up to 37% more torque than the industry reference.
- **Supply Chain Resilience:** Reducing the rare-earth "footprint" by over 30% significantly de-risks production against geopolitical shocks and scarcity.

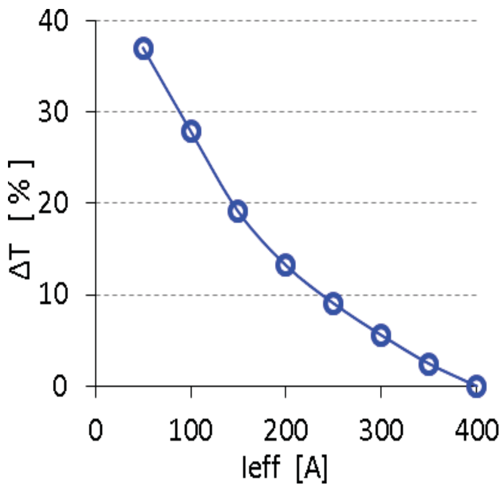

Fig. 4. Torque difference between new and the reference design

FEAAM can assist your team by transforming theoretical physics into market-ready rotor topologies, delivering drivetrains that are lighter, cooler, and more cost-effective.